

What is it?

The chip design phase involves detailing the physical circuit that can implement a particular application on a piece of silicon. The physical circuit would integrate billions of transistors to achieve an application that would suit a particular requirement, such as microprocessor, router or cell phone.

Fabrication involves creating the physical circuit on silicon, and is akin to manufacturing. Assembly involves wrapping the piece of silicon in a package so that it can be used in an electronic circuit.

The early stages of chip design saw the emergence of big players (Intel, AMD, et al.) that did everything on their own, from chip design to fabrication. The second phase saw the rise of Taiwanese stand-alone foundries which enabled the integrated companies to outsource their fabrication functions at a lower cost. This separated the capital-intensive fabrication phase from the people-intensive design phase and allowed countries such as India to get into chip design.

Currently, the key players in the semiconductor design industry can be segregated into:

- ❑ **Fabless firms:** These do only design work and don't undertake actual manufacture of chips
- ❑ **Design service firms:** Those who execute specific contracts for big firms in chip design
- ❑ **Silicon intellectual property firms:** These create IP and then sell it to other firms
- ❑ **EDA (Electronic Design Automation) tool vendors:** Those who evaluate designs for manufacturing readiness

Why is it needed?

The design and manufacture of faster and smaller semiconductors means that they can be used in different applications, thus driving growth of the industry. From being used in mainframe computers in 1970s to PCs in the 1980s to mobile handsets in the 1990s and in consumer electronics at present, the industry has always found a new driver for growth.

The ongoing recession has hit the electronics sector, which, in turn, has taken its toll on semiconductor sales — it is projected that the next generation of application growth will drive from as diverse fields as renewable energy, electric vehicles, healthcare, and so on.

As innovations take place, there will be a need for people who can innovate even further. That is also where specialised chip designers will be preferred over software engineers.

The growth story

India, with its abundant engineering talent and low costs, is an attractive destination, and hence will definitely be an important player in the chip design business.

- A recent ISA-IDC report found that the total chip design market is expected to surpass \$7.37 billion in 2008, with a huge chunk of revenues coming from embedded software, followed by VLSI design and hardware/board design.
- Gartner states that the market for third-party chip design services in India is estimated at about \$400 million. Ganesh Ramamoorthy, Principal Research Analyst, Gartner, says, "Gartner estimates that the market will grow by 26 per cent in 2008, year-on-year. Between 2007 and 2012, we